

Todd Takala

Lead Data Scientist — Lead Machine Learning Engineer

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Summary

Lead Data Scientist delivering over \$1.2 billion in business value. I lead end-to-end data science projects, leveraging machine learning to solve complex challenges, reduce costs, and provide key insights to stakeholders.

Education

Arizona State University

Tempe, AZ

Bachelor of Science in Mechanical Engineering

Graduated: 2006

Solid STEM foundation in statistical modeling and data analysis, with expertise in logic, linear algebra, and finite element analysis. Proven accuracy and willingness to learn in labs focused on mechanisms, materials, and manufacturing.

Professional Experience

Lead Data Scientist

Surprise, AZ

Caterpillar Inc. – cat.com

September 2022 – Present

- Spearheaded the development of a machine learning (ML/AI) system using neural networks and predictive models to forecast equipment repair times, achieving a CEO Award and cutting warranty costs by \$80M in 2024.
- Engineered and deployed robust, end-to-end project management of ETL/MLOps pipelines using Python, Snowflake and AWS (SageMaker, S3, Step Functions, ECR) to ensure automated model retraining and operational excellence.
- Championed MLOps best practices as a steward for GitHub CI/CD, Docker, and AWS architecture, setting new code standards and leading R&D efforts for iterative improvement.
- Led data science projects from stakeholder engagement and KPI setting to development approval, showcasing strong business acumen.
- Designed monitoring plans for models, created Tableau and Power BI dashboards, and used customer feedback to iteratively improve and validate business impact.

Reliability Analytics Manager

Tempe, AZ

Amazon – amazon.com

February 2022 – September 2022

- Established a new reliability benchmark using 2 petabytes of IoT data from 400,000 robots to improve design life metrics.
- Authored an influential white paper, exercising technical communication, and utilized RMarkdown and R Shiny to clarify complex Weibull analysis findings, demonstrating Data Storytelling to leadership.
- Pioneered the successful adoption of the new reliability benchmark by senior leadership through executive influence, establishing a new standard for design evaluation.
- Achieved two-time champion status in Amazon Machine Learning University by designing and optimizing winning regression and recommendation systems, showcasing expertise in machine learning algorithms.

Data Engineering Manager
Mesa, AZ

Empire-Cat – empire-cat.com
January 2021 – February 2022

- MLOps pipeline to deploy AI solutions for the analysis of IoT and telematics data, utilizing machine learning algorithms like ARIMA and SVM for time series forecasting to anticipate demand within the remanufacturing supply chain.
- Data processing and exploration phase by integrating diverse datasets, including condition monitoring and machine component data, to create a robust foundation for predictive models.
- Owned the component exchange program, working directly with clients to forecast component durability and mitigate supply chain disruptions, demonstrating a strong ROI strategy.
- Provided technical mentorship by developing the predictive forecasting models foundational to the component exchange program's success in managing the supply chain.

Reliability Engineering Manager
Mesa, AZ

Empire-Cat – empire-cat.com
January 2012 – January 2021

- Resolved critical design flaws in mining equipment by performing reproducible analysis in R and SQL, resulting in a cost reduction of \$1.2B for clients over six years, showcasing project resilience.
- Created a regression model that identified key factors in equipment performance statistics, leading to a \$62M savings in just 90 days by developing effective data products.
- Directed a high-performance team of up to 8 engineers, fostering an iterative improvement and data-driven approach to solve acute and chronic equipment performance issues for clients through team leadership and cross-functional collaboration.
- Built an ensemble of machine learning models from natural language processing (NLP) analysis of service work order text, failure analyses, parts, and IoT telemetry for continuous optimization of equipment service life and fleet availability.

Core Skills

Redshift, ECR, DBA, ETL, TensorFlow, PyTorch, scikit-learn, transformers, anomaly detection, experiments, pandas, Keras, numpy, git, strategic thinking, adaptability, agile, continuous integration, programming languages, distributed computing, architectural design, statistical tools, data gathering, willing to learn

Certifications

Data Camp: Data Scientist Professional Certificate, Certified SQL Associate, AI Fundamentals Certificate

Canonical: Ubuntu Linux Professional Certificate

GitHub: Career Essentials in GitHub Professional Certificate

Udemy: Streamlit for Snowflake Masterclass

BerkeleyX: The Science of Happiness at Work Professional Certificate

Empire-Cat: Lean 6 Sigma Green Belt

Komatsu North America: Certified Technical Communicator

Society of Automotive Engineers: Cost and Finance for Engineers

Arizona State Board of Technical Registration: EIT, Mechanical Engineering – Registration No. 10670